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Lagged Giant Kelp Canopy Response to Warming and Dissolved Oxygen in the Southern California Channel Islands

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ABSTRACT: Giant kelp (*Macrocystis pyrifera*) forests are highly productive coastal ecosystems, yet many have shown increasing vulnerability to ocean warming and environmental stress. In the California Current System, long-term warming occurs alongside variable coastal upwelling, and warming can also reduce dissolved oxygen, potentially compounding stress on kelp ecosystems. This study examines how sea surface temperature (SST), coastal upwelling, and dissolved oxygen relate to giant kelp canopy dynamics in the Southern California Channel Islands from 1990 to 2024 using publicly available datasets aggregated to a quarterly time scale. I assessed long-term trends in SST and upwelling, tested the relationship between SST and dissolved oxygen, and evaluated whether giant kelp canopy responses to warming occur with a seasonal lag. SST showed a sustained warming signal, while upwelling exhibited strong seasonal and interannual variability without a clear long-term trend. Warmer quarters were associated with lower dissolved oxygen, and giant kelp canopy area displayed interannual variability with worse conditions in recent years. The strongest negative SST–canopy relationship occurred when SST preceded canopy by approximately one quarter, indicating a lag. After accounting for SST, residual oxygen variability remained positively associated with giant kelp canopy area. Overall, the results indicate that ocean warming, dissolved oxygen variability, and giant kelp canopy dynamics are significantly associated over seasonal timescales, with the strongest temperature–canopy relationship occurring after an approximately one-quarter lag.

KEYWORDS: Earth Science, Oceanography, Sea Surface Temperature Variability, Coastal Upwelling Dynamics, Kelp Forest Ecosystem.

■ Introduction

Kelp forests are highly productive coastal ecosystems that provide habitat, food resources, and physical structure for a wide range of marine organisms. As canopy-forming foundation species, kelps strongly influence nearshore community structure, biodiversity, and ecosystem functioning. In recent decades, however, kelp forests have shown increasing vulnerability to ocean warming. Field studies and long-term syntheses have documented declines in kelp abundance, altered recovery dynamics, and reduced ecosystem resilience under elevated temperature conditions.[1,2]

In one of the earliest demonstrations of this pattern, Wernberg *et al.* showed that kelp beds exposed to warmer conditions exhibit reduced resilience, characterized by slower recovery following disturbance and weakened recruitment-driven feedbacks.[1] At broader spatial scales, global analyses have revealed substantial regional variability in kelp forest trends, with many regions experiencing long-term declines associated with climate-driven stressors.[2] Together, these findings suggest that ocean warming may not only reduce kelp abundance but also undermine the capacity of kelp forests to recover from disturbance.

In eastern boundary current systems such as the California Current System (CCS), long-term warming occurs alongside strong seasonal and interannual variability driven by wind-forced coastal upwelling. Upwelling supplies cold, nutrient-rich water to the nearshore environment and can promote kelp growth, yet its intensity and timing vary considerably across space and time.[3] Physical oceanographic studies show that SST patterns in the CCS are structured by circulation features and temperature fronts that shape local thermal conditions and biological responses.[3,4] While upwelling can temporarily reduce surface temperatures, it remains unclear whether this variability can offset persistent warming trends, particularly during extreme climate events.

Recent marine heatwaves have intensified concerns about kelp forest vulnerability in the CCS. Di Lorenzo and Mantua identified the 2014–2015 North Pacific marine heatwave as a multi-year warming event that produced persistent physical anomalies across the region.[5] Subsequent ecological analyses showed that kelp forest communities along the California Current exhibited strong, but geographically variable, responses to this event, with southern regions often displaying greater sensitivity to warming.[6] These observations highlight the importance of examining kelp responses within a regional context and over sufficiently long timescales.

Ocean warming also alters the chemical environment of coastal waters. As temperature increases, dissolved oxygen concentrations generally decline due to reduced solubility and enhanced stratification, which limits vertical mixing and ventilation. Keeling *et al.* identified ocean deoxygenation as a major consequence of climate change, with significant implications for marine ecosystem functioning.[7] Experimental studies further demonstrate that biological responses to warming often reflect the combined effects of temperature, oxygen, and other environmental stressors rather than temperature alone.[8] Within kelp forest ecosystems, reduced oxygen availability may influence many associated organisms, particularly animals that

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rely on dissolved oxygen for respiration. The direct effects of reduced dissolved oxygen on giant kelp (*Macrocystis pyrifera*) are less well understood, but dissolved oxygen remains a useful indicator of broader environmental conditions associated with ocean warming and stratification.

Importantly, kelp forest responses to environmental stressors are not always immediate. Laboratory experiments by Weigel *et al.* demonstrated that elevated temperatures can strongly reduce survival and recruitment success in microscopic kelp life stages, even when adult canopies remain temporarily intact.[9] Field-based research similarly indicates that warming can erode kelp resilience by slowing post-disturbance recovery, leading to delayed declines in canopy cover.[1] These findings suggest that kelp forest responses to climate forcing may occur on seasonal timescales, emphasizing the importance of accounting for temporal lags when linking physical drivers to observed changes in giant kelp canopy structure.

The Southern California Channel Islands represent a transitional region within the CCS where kelp forests are influenced by long-term warming, episodic upwelling, and recent marine heatwaves. Despite growing recognition of climate-driven kelp decline, the combined roles of SST trends, dissolved oxygen variability, and upwelling forcing in shaping giant kelp canopy dynamics in this region remain incompletely resolved. Multi-decadal, integrative analyses are therefore needed to disentangle these interacting drivers and clarify the mechanisms linking climate variability to kelp forest persistence.

Previous studies in southern California have identified temperature, nutrient availability, disturbance, and large-scale climate variability as important drivers of giant kelp (*Macrocystis pyrifera*) dynamics. Long-term analyses using field observations and remote sensing datasets have shown that environmental conditions strongly influence giant kelp biomass, canopy persistence, and synchrony across regions.[10-13] In this study, I use long-term public datasets to examine how SST, coastal upwelling, dissolved oxygen, and giant kelp (*Macrocystis pyrifera*) canopy dynamics covary in the Southern California Channel Islands from 1990 to 2024.[14-17] Rather than assuming a direct causal effect of dissolved oxygen on giant kelp canopy area, this study evaluates dissolved oxygen as a biogeochemical indicator associated with ocean warming, stratification, and broader environmental conditions that may influence giant kelp canopy dynamics. Unlike many studies that focus primarily on warming impacts or community responses over shorter windows, this analysis integrates temperature, upwelling forcing, dissolved oxygen, and satellite-based canopy observations across multiple decades in one consistent framework.[7,18] The research question guiding this study is: *How do changes in SST and upwelling variability influence dissolved oxygen, and how do these changes relate to giant kelp canopy area, including delayed responses?*

Based on prior work, this study evaluates three predictions: (1) SST is negatively associated with dissolved oxygen, confirming the expected temperature–oxygen relationship in the study region; (2) giant kelp canopy area is most strongly (and negatively) related to SST when SST is lagged by approximately one quarter; and (3) after accounting for SST, residual oxygen variability remains associated with giant kelp canopy area.[1,7]

Methods

Study Area and Time Frame:

This study examines oceanographic and ecological variability across the five northern Channel Islands (Anacapa, Santa Cruz, Santa Rosa, San Miguel, and Santa Barbara Islands), which comprise Channel Islands National Park in Southern California. This region is strongly influenced by the California Current System (CCS) and seasonal wind-driven coastal upwelling, creating substantial variability in temperature, nutrient availability, and biological productivity. Giant kelp forests in this region are shaped by the interaction of long-term ocean warming, episodic upwelling, and extreme climate events, making it a suitable system for investigating climate–ecosystem linkages.[1,6,10-13] The analysis spans January 1990 through December 2024, a period that captures multi-decadal climate trends as well as recent marine heatwaves, including the 2014–2015 warming event.[5]

Because the datasets used in this study originate from multiple sources with different native spatial and temporal resolutions, all variables were aggregated to a quarterly time scale. Quarterly aggregation captures seasonal to interannual variability while reducing short-term noise associated with monthly fluctuations, and is consistent with the timescales at which kelp forest responses to environmental stress have been shown to occur.[1,9] Spatial resolution was not standardized because each dataset represents environmental conditions at its native observational scale. SST and dissolved oxygen products provide regional oceanographic conditions, the Bakun Upwelling Index represents a regional atmospheric forcing metric, and the giant kelp canopy dataset provides high-resolution satellite observations of giant kelp. Aggregating all datasets to a common spatial resolution would reduce the spatial detail of the kelp observations without improving comparability among variables, as the analyses focus on regional temporal relationships rather than pixel-by-pixel spatial comparisons.

Sea Surface Temperature:

Sea surface temperature (SST) data were obtained from the NOAA Extended Reconstructed Sea Surface Temperature version 5 (ERSSTv5) dataset.[14] ERSSTv5 provides a global, monthly reconstruction of SST based on in situ ship and buoy observations that are statistically interpolated and bias-corrected to reduce systematic measurement errors.

Monthly SST values (°C) were extracted for the surface layer (0 m depth) within the region bounded by 32°N–35°N latitude and 239°E–242°E longitude (equivalent to 121°W–118°W). This spatial domain encompasses the Southern California Channel Islands and surrounding coastal waters. The dataset has a native spatial resolution of 2° × 2°. SST data from 1990 to 2024 were averaged into quarterly means and used as an indicator of long-term ocean warming, consistent with prior studies examining temperature-driven giant kelp (Macrocystis pyrifera) responses in the CCS.[1,2,10] Although ERSSTv5 has a relatively coarse spatial resolution (2° × 2°) and does not

resolve fine-scale thermal gradients among individual Channel Islands, it provides a consistent multi-decadal SST record spanning the full study period and was therefore selected to evaluate long-term regional temporal variability rather than local spatial variability.

Coastal Upwelling:

Coastal upwelling was characterized using the Bakun upwelling index obtained from the NOAA Physical Sciences Laboratory (PSL) monthly time-series archive.[15] The analysis used the standard Bakun Upwelling Index calculated at 33°N, 119°W, a location offshore of the Southern California Channel Islands that was selected to represent regional wind-driven coastal upwelling conditions within the study area.

The Bakun index estimates offshore Ekman transport driven by alongshore wind stress and is reported in units of m^3 s^{-1} per 100 m of coastline, with positive values indicating upwelling-favorable conditions. Monthly index values spanning 1967–2024 were downloaded and restricted to the 1990–2024 study period. Values were averaged into quarterly means to match the temporal resolution of the other datasets. Upwelling was included as a physical forcing variable because it influences SST, nutrient supply, and nearshore environmental conditions within the California Current System.[3,4,11,19] Because the objective of this study was to examine regional temporal relationships rather than local spatial variability, a single representative Bakun index was used to characterize upwelling conditions across the study region.

Dissolved Oxygen:

Dissolved oxygen data were obtained from the Global Ocean Biogeochemistry Hindcast produced by the Copernicus Marine Environment Monitoring Service (CMEMS).[16] Specifically, I used the monthly dataset *cmems_mod_glo_bgc_my_0.25deg_P1M-m* from the GLOBAL_MULTIYEAR_BGC_001_029 product, which integrates ocean circulation models with biogeochemical components constrained by satellite and in situ observations. The dataset has a native horizontal spatial resolution of approximately 0.25° × 0.25°.

Monthly mean dissolved molecular oxygen concentrations (mmol m^{-3}) were extracted for the region bounded by 32°N–35°N latitude and 121°W–118°W longitude. To approximate oxygen conditions relevant to kelp forest habitats, values were vertically averaged over the upper 0.51–9.82 m of the water column. Dissolved oxygen data were available from 1993 onward; therefore, analyses involving oxygen were limited to the 1993–2024 period. Quarterly means were calculated for consistency with other variables. Dissolved oxygen was included because it is closely coupled with temperature and broader oceanographic conditions under climate warming, making it a useful environmental indicator within the study region.[7]

Kelp Canopy Area:

Giant kelp canopy data were obtained from the Santa Barbara Coastal Long Term Ecological Research (SBC LTER) program and archived through the Environmental Data Initiative (EDI) under package ID *knb-lter-sbc.74.31*.[17] This dataset, developed by Bell *et al.*, provides a long-term record of giant kelp (*Macrocystis pyrifera*) canopy area and biomass derived from multispectral satellite imagery collected by Landsat 5, Landsat 7, Landsat 8, and, more recently, Landsat 9. The Landsat imagery has a native spatial resolution of 30 m × 30 m.

At the Channel Islands, the Landsat time series represents the canopy area and biomass of giant kelp (Macrocystis pyrifera). The data are provided as quarterly NetCDF files and cover the California coastline, including the Channel Islands. For this study, giant kelp canopy area from 1990 to 2024 was extracted and used as a measure of giant kelp canopy dynamics, consistent with previous remote sensing studies of giant kelp canopy variability in California.

Data Processing:

All data processing and statistical analyses were conducted in R using RStudio. Time stamps were standardized and aligned by calendar quarter. Missing data were minimal for SST and upwelling. Dissolved oxygen data were unavailable before 1993; therefore, analyses involving oxygen were restricted to periods of overlapping data availability.

To facilitate comparisons among variables with different units and magnitudes, selected time series were standardized using z-scores. Standardization was applied for visualization and comparative analysis and did not alter the underlying statistical relationships among variables.

Statistical Analysis:

Long-term variability in SST, coastal upwelling, dissolved oxygen, and giant kelp canopy area was examined using time-series analysis and linear trend assessment. Relationships among variables were evaluated using Pearson correlation coefficients.

To account for delayed biological responses documented in kelp forests,[1,9] lagged correlations were calculated between SST and kelp canopy area using multi-quarter lags. In addition, to evaluate the role of dissolved oxygen independent of temperature, dissolved oxygen was regressed against SST, and the resulting residuals were correlated with giant kelp canopy area. This approach allowed oxygen variability to be examined separately from the overall temperature trend.

All figures were produced using *ggplot2* in R and include time-series plots, scatterplots with fitted regression lines, lagged relationship plots, and multi-panel figures. All datasets used in this study are publicly available, and the analysis workflow was designed to ensure transparency and reproducibility.

■ Results and Discussion

This section reports results in the same sequence as the research question: (i) establish the physical forcing context, (ii) describe kelp canopy variability, and (iii) test linkages among SST, dissolved oxygen (DO), and kelp canopy dynamics, including delayed (lagged) responses. All variables were analyzed at a quarterly time step using publicly available datasets (SST: ERSSTv5;[14] upwelling: Bakun Index;[15] DO: Copernicus hindcast;[16] kelp canopy: SBC LTER Landsat product).[17]

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Conceptual framework and hypotheses:

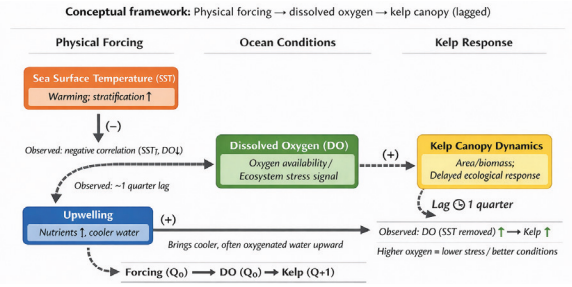


Figure 1: Conceptual framework linking physical forcing, ocean conditions, and lagged giant kelp canopy response. This framework illustrates the hypothesized pathway tested in this study: warming SST is associated with lower dissolved oxygen, and the giant kelp canopy responds to these changes with a seasonal delay rather than immediately. The arrows represent hypothesized relationships evaluated in this study and do not imply direct causation.

Figure 1 summarizes the conceptual framework examined in this study. Physical forcing, including warming SST and variable coastal upwelling, influences regional oceanographic conditions, including dissolved oxygen, while giant kelp (*Macrocystis pyrifera*) canopy dynamics are evaluated in relation to these environmental variables with explicit consideration of seasonal response lags. The conceptual framework is intended to illustrate the relationships evaluated in this study and should not be interpreted as demonstrating a direct causal effect of dissolved oxygen on giant kelp canopy dynamics. Rather, dissolved oxygen is examined as a biogeochemical indicator that covaries with ocean warming and broader environmental conditions. The research question guiding this analysis is: *How do changes in SST and upwelling variability influence dissolved oxygen, and how do these changes relate to kelp canopy area, including delayed responses?* This framing is consistent with prior work showing that giant kelp is sensitive to warming-related environmental stress and that ocean warming is associated with declining dissolved oxygen concentrations in marine systems.[7,10]

Trends and variability in physical forcing:

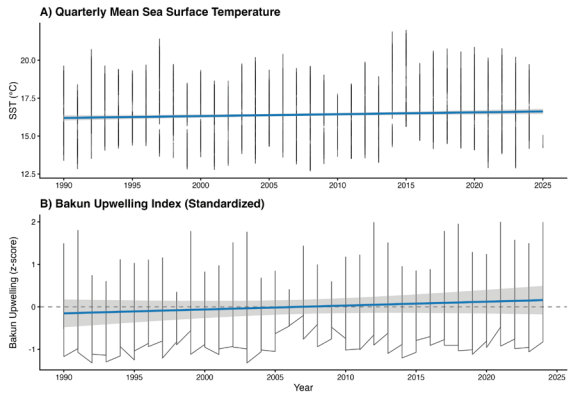


Figure 2: Trends and variability in physical forcing in the Southern California Channel Islands from 1990–2024. (A) Quarterly mean sea surface temperature (SST). Vertical lines indicate the range of quarterly SST values observed within each year, and the blue line represents the fitted linear trend. (B) Standardized Bakun Upwelling Index. Vertical lines indicate the range of quarterly standardized upwelling values within each year, the blue line represents the fitted linear trend, and the shaded region denotes the 95% confidence interval of the fitted trend.

Figure 2 summarizes long-term patterns in sea surface temperature (SST) and coastal upwelling from 1990 to 2024. Quarterly mean SST exhibited substantial within-year variability while showing an overall increasing trend through time (Figure 2A), consistent with long-term ocean warming in the California Current System. In contrast, the standardized Bakun Upwelling Index fluctuated considerably among years without a consistent long-term increase or decrease (Figure 2B), reflecting the strong seasonal and interannual variability characteristic of coastal upwelling in the region. These contrasting patterns suggest that warming has occurred despite persistent variability in wind-driven upwelling.

Temporal patterns in kelp canopy:

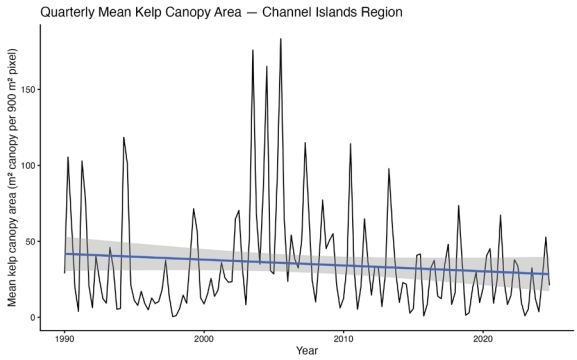


Figure 3: Quarterly mean giant kelp (*Macrocystis pyrifera*) canopy area in the Channel Islands region from 1990 to 2024. The blue line represents the fitted linear trend, and the shaded region indicates the 95% confidence interval of the fitted trend. Giant kelp canopy exhibits pronounced boom–bust dynamics with high interannual variability throughout the study period.

Figure 3 shows that the giant kelp (*Macrocystis pyrifera*) canopy area was highly episodic throughout the study period, with sharp increases followed by rapid declines and prolonged periods of relatively low canopy area. The fitted trend line indicates a slight long-term decline in average canopy area, although the dominant pattern is substantial interannual variability rather than a smooth directional trend. Similar boom–bust dynamics and heterogeneous long-term changes have been reported in previous studies of giant kelp dynamics in California.[1,2,10-13]

Dissolved oxygen response to temperature:

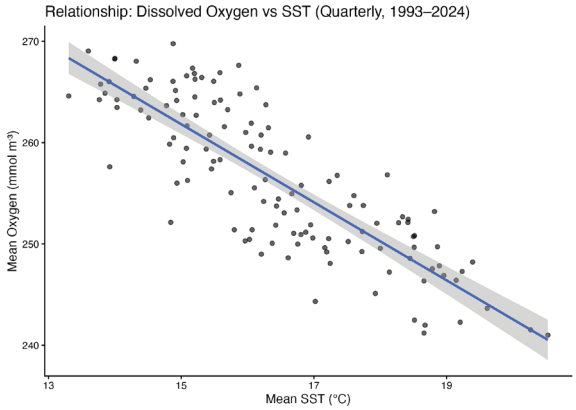


Figure 4: Relationship between SST and dissolved oxygen (quarterly, 1993–2024). Dissolved oxygen declines steeply with increasing SST, demonstrating a strong and statistically significant negative temperature–oxygen coupling consistent with climate-driven deoxygenation.

Figure 4 tests the temperature–oxygen pathway during the overlapping DO record (1993–2024). DO declines strongly as SST increases, and the fitted relationship is steep and highly significant (slope = −3.85 mmol m^{-3} per °C, p = 3.3 × 10^{-34}). Mechanistically, this pattern is consistent with reduced oxygen solubility in warmer water and stratification limiting ventilation, which are key components of deoxygenation under warming.[7] This relationship is consistent with the expected physical coupling between sea surface temperature and dissolved oxygen under ocean warming and establishes the temperature–oxygen relationship examined in this study. This result supports Prediction (1): SST is negatively associated with dissolved oxygen.

Seasonal lag in kelp response to warming:

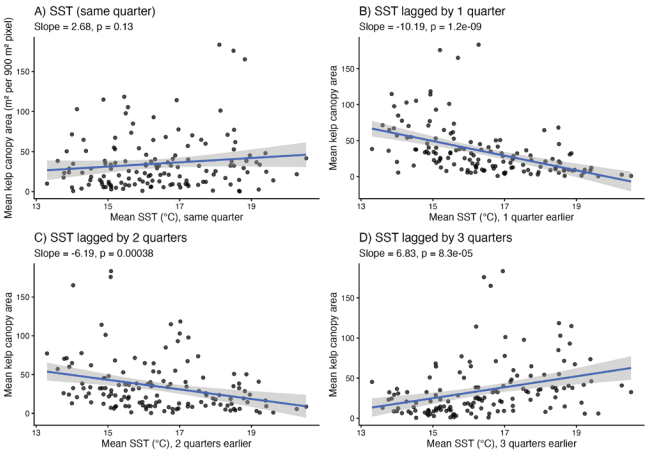


Figure 5: Kelp canopy area vs SST across temporal lags (0–3 quarters). The strongest negative relationship occurs when SST is lagged by one quarter, indicating that giant kelp canopy decline most clearly follows warming with a seasonal delay rather than within the same quarter.

Figure 5 evaluates whether the giant kelp canopy responds to SST in the same quarter or with a delay. When SST and giant kelp canopy area are measured in the same quarter (Figure 5A), the association is weak and not significant (slope = 2.88, p = 0.13). In contrast, SST lagged by one quarter (Figure 5B) shows the strongest negative relationship (slope = −10.1, p = 1.2 × 10^{-9}), indicating that warmer conditions are followed by reduced canopy in the subsequent season. The relationship remains negative but weaker at a two-quarter lag (Figure 5C; slope = −6.19, p = 3.8 × 10^{-4}), and reverses by three quarters (Figure 5D; slope = 6.83, p = 3.0 × 10^{-5}), suggesting that the most interpretable temperature signal is concentrated within roughly one seasonal cycle.

Table 1 quantifies these differences across lags. The same-quarter model explains almost none of the variance in canopy area (R^2 = 0.016; Adj. R^2 = 0.009; p = 0.13), while the one-quarter lag model is both the strongest and the best-fitting relationship (β = −10.187 ± 1.562, t = −6.520, p = 1.25 × 10^{-9}; R^2 = 0.237; Adj. R^2 = 0.231; n = 139). The two-quarter lag remains negative but is notably weaker (β = −6.187 ± 1.697; R^2 = 0.089; p = 3.78 × 10^{-4}), and by three quarters the sign reverses to a positive association (β = 6.827 ± 1.682; R^2 = 0.109; p = 8.30 × 10^{-5}). At four quarters, the effect is smaller and only modestly significant (β = 4.557 ± 1.745; R^2 = 0.048; p = 0.010), reinforcing that the clearest temperature signal occurs within roughly one seasonal cycle.

This one-quarter lag is ecologically reasonable because canopy loss can reflect accumulated thermal stress that becomes apparent after a warm period rather than within the same quarter. Previous studies have shown that warming can reduce giant kelp resilience and slow recovery following disturbance, supporting the interpretation that canopy responses may lag behind changes in environmental conditions.[1] Together, these results support Prediction (2): giant kelp (*Macrocystis pyrifera*) canopy area is most strongly (and negatively) related to SST at an approximately one-quarter lag.

Table 1: Linear models relating giant kelp canopy area to SST at 0–4 quarter lags. Model results confirm that the one-quarter lag produces the strongest negative association and highest explanatory power, supporting a delayed kelp response to warming.

	Model	Beta	SE	t	p	R2	AdjR2	n
1	SST (same quarter)	2.677	1.767	1.515	1.32e-01	0.016	0.009	140
2	SST lag 1 quarter	-10.187	1.562	-6.520	1.25e-09	0.237	0.231	139
3	SST lag 2 quarters	-6.187	1.697	-3.646	3.78e-04	0.089	0.082	138
4	SST lag 3 quarters	6.827	1.682	4.059	8.30e-05	0.109	0.102	137
5	SST lag 4 quarters	4.557	1.745	2.611	1.01e-02	0.048	0.041	136

Independent role of oxygen availability:

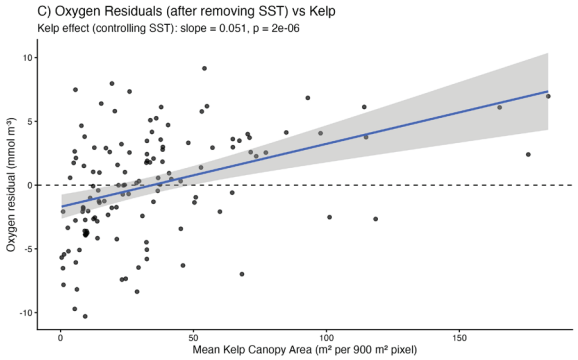


Figure 6: Relationship between SST-controlled oxygen residuals and giant kelp (*Macrocystis pyrifera*) canopy area. After accounting for temperature, higher dissolved oxygen residuals remain positively associated with giant kelp canopy area. This relationship represents a statistical association and should not be interpreted as evidence of a direct causal effect of dissolved oxygen on giant kelp canopy dynamics.

To isolate oxygen variation not explained by temperature, DO was regressed on SST, and residuals were calculated (i.e., "SST-controlled oxygen"). Figure 6 shows a significant positive association between these oxygen residuals and giant kelp (*Macrocystis pyrifera*) canopy area (slope = 0.051, p = 2.0 × 10^{-6}). In other words, for a given temperature background, quarters with relatively higher oxygen tend to have more giant kelp canopy. This relationship should not be interpreted as evidence of a direct causal effect of dissolved oxygen on the giant kelp canopy. Rather, it indicates that dissolved oxygen variability covaries with giant kelp canopy after accounting for SST and may reflect broader environmental conditions associated with giant kelp dynamics. Because dissolved oxygen also covaries with other environmental variables, including nutrient availability, the observed association may reflect broader oceanographic conditions rather than an independent effect of dissolved oxygen. Future studies incorporating nitrate ob-

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servations could help distinguish the relative contributions of dissolved oxygen and nutrient availability to giant kelp canopy dynamics. This result supports Prediction (3): SST-controlled oxygen variability remains positively associated with giant kelp canopy area.

Integrated interpretation:

Taken together, these results answer the research question by showing that warming SST is associated with reduced oxygen availability, and that kelp canopy decline is most strongly linked to SST after a seasonal delay, rather than within the same quarter. Upwelling remains highly variable through time, indicating that short-term physical variability continues to shape local conditions, but does not remove the long-term warming background captured in SST. Finally, oxygen conditions remained significantly associated with giant kelp canopy area after accounting for SST, suggesting that dissolved oxygen covaries with other environmental conditions relevant to giant kelp dynamics rather than demonstrating an independent causal effect.

Conclusion

This study combined multi-decadal public datasets to evaluate how sea surface temperature (SST), coastal upwelling variability, and dissolved oxygen relate to kelp canopy dynamics in the Southern California Channel Islands from 1990 to 2024.[14-17] The record indicates sustained warming in SST alongside strong seasonal and interannual variability in upwelling, while giant kelp canopy exhibited pronounced boom–bust dynamics with an overall shift toward lower canopy conditions in more recent years, consistent with reduced resilience under repeated climate stress.[1,10] Warmer quarters were associated with lower dissolved oxygen, supporting the expected temperature–oxygen coupling under climate change.[7] Importantly, the kelp response to temperature was delayed rather than immediate: the clearest SST–canopy relationship emerged when SST preceded canopy by roughly one quarter, consistent with seasonal-scale processes such as accumulated physiological stress and recruitment limitation.[1,9] Even after accounting for SST, residual oxygen variability remained positively associated with giant kelp canopy area. This relationship should not be interpreted as evidence of a direct causal effect of dissolved oxygen but rather as an association that may reflect broader environmental conditions. Overall, these findings indicate that warmer SST was associated with lower dissolved oxygen, that giant kelp canopy declines were most strongly associated with SST after an approximately one-quarter lag, and that relatively higher dissolved oxygen residuals were associated with greater giant kelp canopy area.

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